

Xinchen Yao

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Education

Dual degree program in Zhejiang University - University of Illinois Urbana-Champaign Institute:

University of Illinois Urbana-Champaign, Urbana, Illinois, US

Sep 2022 – May 2026

BS in Computer Engineering, GPA: 3.77/4.0

Zhejiang University, Zhejiang, China

Sep 2022 – May 2026

BS in Electronic and Computer Engineering, GPA: 3.83/4.3

Awards and Honors

- Outstanding Student in Academic Achievement, ZJU-UIUC Institute (2024 - 2025)
- Zhejiang University Scholarship (2024 - 2025)
- Second Prize in RoboMaster¹ University League Competition (2023, 2024)

Experience

Physical Intelligence Lab², Zhejiang, China

Undergraduate Researcher

Zhejiang University - University of Illinois Urbana-Champaign Institute

Jul 2025 - Present

Advisor: **Prof. Hua Chen³**

- Developing locomotion controllers for bipedal robots and humanoids using reinforcement learning.
- Building evaluation pipelines to test trained robot policies across multiple simulation environments and deploying them to physical robots.
- Researching methods to reduce the performance gap between simulated and real-world robot operation through learned dynamics models.
- Improving reinforcement learning algorithms to achieve better training efficiency and performance stability.

Human Dynamics and Controls Lab⁴, Illinois, US

Undergraduate Researcher

The Grainger College of Engineering, University of Illinois Urbana-Champaign

Sep 2024 - Jun 2025

Advisor: **Prof. Elizabeth Hsiao-Wecksler⁵**

- Developed a dynamics model for a self-balancing ballbot platform, simulating omniwheel behavior while maintaining computational efficiency.
- Implemented learning-based control to optimize balancing and movement performance and robustness.
- Integrated new force sensors into robot hardware, improving state estimation and control reliability.

RoboMaster Meta Team⁶, Zhejiang, China

Control Group Leader

Zhejiang University - University of Illinois Urbana-Champaign Institute

Jun 2023 - Present

Advisor: **Prof. Jiahuang Cui⁷**

- Led control group to win second prize in RoboMaster University League competition, managing a team developing software for multiple competitive robots.
- Designed and implemented a distributed control system supporting real-time coordination for multiple robots with modular architecture.
- Developed software for motor control, sensor integration, autonomous navigation, and decision algorithms.
- Trained and mentored new team members on control theory, robot software development, and algorithm design.
- Collaborated with mechanical team to optimize robot design for improved controllability and performance.

¹<https://www.robomaster.com/en-US>

²<https://physicalintelligence-lab.github.io/>

³<https://zjui.intl.zju.edu.cn/en/team/teacherinfo/2461>

⁴<https://hdcl.mechanical.illinois.edu>

⁵<https://bioengineering.illinois.edu/people/ethw>

⁶<https://github.com/Meta-Team>

⁷<https://mechse.illinois.edu/people/profile/zjui-cui>

Technical Skills

Programming: Python, C/C++, CUDA, Matlab, Rust

Machine Learning: PyTorch, Jax, Reinforcement Learning, Generative Models

Robotics: Robot Simulators, Control Theory, ROS/ROS2, Computer Vision, SLAM

Systems: Linux, Embedded Systems, Parallel Computing

Projects

Where to Learn: Analytical Policy Gradient Directed Exploration for On-Policy Robotic Reinforcement Learning (under review by IEEE RA-L)

Second Author

- Implemented a novel reinforcement learning algorithm in Brax that enhances exploration efficiency and training stability for robotic control.
- Evaluated the algorithm in Mujoco Playground across multiple benchmark environments with comprehensive hyperparameter tuning and ablation studies.
- Successfully trained and deployed a walking policy on a physical biped robot for real-world validation.
- Website: <https://wheretolearn.github.io>

Omni WBR: Learning Adaptive Hybrid Wheeled-Biped Robot Omnidirectional Locomotion via Position-Based Incentive (in progress)

First Author

- Developed a task design to generate adaptive omnidirectional gaits for wheeled bipedal robots, enabling smooth movement on uneven terrains without explicit gait planning.
- Implemented a training and deploying pipeline to evaluate the method in Isaac Lab simulator and physical robot.
- Demonstration available on my website: <https://yao-xinchen.github.io/projects/omni-wbr>

Meta-Team/Meta-ROS: Modular ROS2 Control Framework for multiple RoboMaster Competition Robots

Creator, Maintainer

- Created a comprehensive robot control system in ROS2 supporting multiple competitive robots with sensor integration, motion control, and autonomous coordination.
- Designed modular architecture enabling easy extension and real-time performance for competition scenarios.
- Integrated advanced capabilities including visual recognition, trajectory prediction, localization, and learning-based controllers.
- Code: <https://github.com/Meta-Team/Meta-ROS>

Custom Wheeled Biped Robot with RL-base Locomotion Controller

Leader

- Co-designed and built a custom wheeled biped robot with 4 degrees of freedom and closed chain structure.
- Trained a reinforcement learning-based locomotion controller and deployed it on the physical robot.
- Demonstration available on my website: <https://yao-xinchen.github.io/projects/wheeled-biped>

Course Projects

Unix-Like RSIC-V Operating System, Including Kernel and Shell

ECE391 Computer Systems

- Implemented core operating system components including device drivers, filesystem, virtual memory, process management, and multitasking capabilities.
- Designed and implemented a Unix-style shell with process spawning and management.
- Composed a tutorial on workflow setup: <https://github.com/Yao-Xinchen/ECE391-Workflow>

Autonomous Diagonal and Parallel Parking

ECE484 Safe Autonomy

- Implemented localization and mapping system, and visual path recognition using deep learning.
- Designed and implemented autonomous parking system with diagonal and parallel parking maneuvers.
- Demonstration available on my website: <https://yao-xinchen.github.io/projects/auto-parking>